

Valora: A Human-Centric Digital Currency for Universal Financial Freedom

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Valora is a decentralized digital currency that establishes Proof-of-Personhood (PoP) as its core consensus mechanism, ensuring that every unique human obtains an equal voting share in block production and validation. Unlike Proof-of-Work or Proof-of-Stake protocols, which naturally concentrate power in the hands of resourceful miners or large stakeholders (Gervais, Ritzdorf, Karame, & Capkun, 2016; Nakamoto, 2008), Valora precludes such monopolies by anchoring governance rights to verifiable human identities. It enforces privacy by default, leveraging advanced cryptography to shield all transactional data, and employs a minimal, UTXO-based ledger to streamline security and preserve performance. The overarching goal is to create a censorship-resistant payment system that just works for global users, including those in low-connectivity or politically restricted settings (Androulaki, Karame, Roeschlin, Scherer, & Capkun, 2013).

The protocol’s principal innovation is its reliance on PoP to allocate one credential per real human participant (Baum, 2016). This approach obviates the “one-CPU–one-vote” or “one-stake–one-vote” paradigms that often result in resource- or wealth-based centralization (Badev & Chen, 2014). PoP ceremonies, which can be in-person or online Turing tests, verify users without demanding personal details on-chain. The system admits only those credentials that meet concurrency checks (no individual can appear at multiple ceremonies simultaneously), making identity infiltration far more difficult than simply accumulating hashing power or tokens. By using multiple independent communities to host ceremonies, Valora distributes trust away from a single agency. The blockchain converges on which identity proofs to accept only when no contradictions arise. Should someone attempt to produce thousands of fake identities, they must circumvent these concurrency requirements, a task orders of magnitude harder than controlling mining pools.

Leader selection for block proposals proceeds by randomly sampling from this PoP-verified pool, ensuring every identity is chosen roughly in proportion to its single share (Lenstra & Wesolowski, 2017). Over repeated intervals, block production thus remains equitably distributed. Once a proposer is designated,

it references the previous block’s hash, compiles valid transactions, signs the block header with its private key, and broadcasts it. Peers verify that the block belongs to an authentic identity credential and that its transactions meet UTXO correctness rules. The resulting structure strongly mitigates the risk of a 51% attack (Nakamoto, 2008) because compromising the chain requires subverting over half of globally recognized participants, a challenge far more demanding than amassing specialized hardware in Proof-of-Work.

Valora’s network layer is a decentralized mesh that relies on gossip protocols and possible fallback communication channels like SMS or satellite broadcasts (Decker & Wattenhofer, 2013). Nodes discover each other via seed addresses or DHT-based lookups, forming an unstructured overlay. Traffic is encrypted to hamper censorship and packet inspection, and users in high-restriction regions can connect through anonymity networks like Tor (Dingledine, Mathewson, & Syverson, 2004). The system thereby maintains connectivity even under aggressive interference, since adversaries must block an unfeasibly large fraction of global nodes to silence the ledger.

Valora’s ledger design follows a UTXO structure that is conducive to both privacy and parallel verification (Karame, Androuraki, & Capkun, 2012). Each transaction cites one or more outputs as inputs, proving ownership cryptographically, and creates new outputs assigned to fresh addresses. Because transactions specify discrete references, nodes can validate them concurrently without touching a global account state. Blocks need not carry arbitrary data: the protocol disallows embedding additional information in outputs or transaction fields beyond what is essential for currency functionality. Historical investigations into blockchain usage indicate that arbitrary data, if permitted, leads to ledger bloat and tangential or even problematic payloads that cannot easily be pruned (Navarro-Arribas, 2018). By prohibiting free-form on-chain data, Valora preserves node accessibility and avoids potential liabilities.

Privacy by default is another principal commitment, satisfied through ring signatures, stealth addresses, and range proofs. Ring signatures obscure which input key in a set of decoys is genuinely being spent (Rivest, Shamir, & Tauman, 2001). Stealth addresses allocate ephemeral one-time addresses to recipients, preventing outsiders from mapping transactions to individuals (Abe, Ohkubo, & Suzuki, 2002). Range proofs, such as Bulletproofs (Bünz, Bootle, Boneh, Poelstra, Wuille, & Maxwell, 2018), or zero-knowledge proofs (Groth, 2016) conceal amounts while permitting nodes to confirm that no hidden inflation occurs. This design aligns with the robust anonymity model previously explored by systems like Monero or Zcash (Fujisaki & Suzuki, 2007). The ledger itself stores only obfuscated references: it is infeasible for an external observer to determine which inputs belong to which outputs, or how much value is transferred. The result is an effectively fungible coin supply that resists blacklists or forced reidentification.

The architecture supports fast block intervals (e.g., around one minute or less) since there is no computational race to generate proofs-of-work. Although rapid

intervals can increase potential forks, the randomness-based consensus limits block proposer collisions, and optionally, a committee of random identities can quickly finalize each block (Castro & Liskov, 2002). This approach substantially improves throughput and confirmation times relative to Bitcoin (Nakamoto, 2008). If load grows further, participants can rely on off-chain channels or rollups (Buterin, 2018) to batch many transactions, leaving the main ledger for settlement and high-value operations. Sharding could also be explored, distributing the UTXOs and identity sets across multiple shards so that cross-shard communication references a single global PoP registry (Karame et al., 2012). These adaptations ensure that Valora remains decentralized even under high usage.

Valora’s security model addresses classical threats: Sybil attacks are blocked by concurrency-limited PoP verification, 51% infiltration is rendered nearly impossible by requiring a majority of the global user base, censorship is undermined by private transactions and unstructured gossip, and denial-of-service attempts are constrained by transaction fees plus local rate limiting (Boneh & Shoup, 2020). Governance is likewise PoP-based: critical updates, such as ring signature parameter changes or inflation tuning, must pass a one-human-one-vote process that cannot be swayed merely by holding tokens (Buterin, 2018). Such governance is less susceptible to wealthy coalitions or miner groups forcibly altering protocol rules.

The monetary policy, reflecting the system’s egalitarian ethos, mints a fixed initial supply (ten billion “Vals”) and distributes a set allotment (e.g., one thousand Vals) to each newly verified identity. Should user adoption exceed that supply, the chain triggers a mild inflation (e.g., one to two percent annually), which preserves equal rights for latecomers without generating unchecked dilution (Badev & Chen, 2014). The minimal transaction unit is one Val, so typical everyday payments might simply be denominated in single integer units. If sub-Val denominations become necessary, PoP-governed proposals can add them. This balance between convenience and identity-based fairness aims to improve upon Bitcoin’s halving schedule, which can disadvantage late adopters (Nakamoto, 2008), and to avoid resource-based reward structures that concentrate economic power among a handful of industrial participants.

Production readiness demands thorough cryptographic audits and concurrency-safe node implementations. While this paper refrains from focusing on one specific programming language, it does require memory-safe development frameworks for the node software, concurrency primitives to handle multiple peer connections, and well-reviewed cryptographic libraries to handle ring signatures or zero-knowledge proofs (Groth, 2016). Each client must handle the entire chain verification, from PoP identity checks to private transaction logic, ensuring that every node can detect invalid states early. Multi-client diversity is strongly encouraged, mitigating the risk of a single software exploit (Decker & Wattenhofer, 2013).

In essence, Valora builds on the lessons of Bitcoin yet corrects many of its

structural shortcomings: it eliminates the arms race for hashpower, integrates privacy at the protocol level rather than as an afterthought, and places governance in the hands of unique individuals rather than large capital holders. Coupled with layer-2 expansions and flexible design for future cryptographic upgrades, Valora aspires to realize an equitable, decentralized, and censorship-resistant global currency that truly serves the entire population. By insisting on human identity as the primary scarce resource and guaranteeing anonymity for all ordinary transactions, Valora seeks to rekindle the original promise of digital cash: a neutral, user-protective financial infrastructure accessible to all, even under the most constraining socio-political conditions.

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